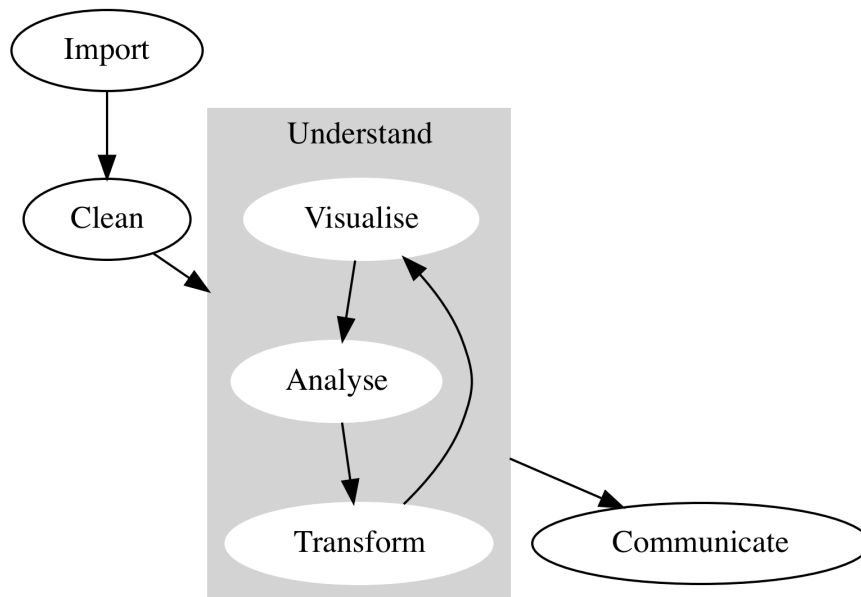


8 Homework Assignment

Due: Jan 23, 2026



Problem Description

Spatial data are inherently linked to the Earth's curved surface, yet most analyses and visualizations occur in two-dimensional Cartesian space. Map projections provide a bridge between these representations, but at the cost of distortion. This assignment examines how different coordinate reference systems distort area, distance, shape, and direction, and how those distortions affect analytical conclusions. Implement your analysis in R or Python, using proper data and multiple map projections, and quantitatively assess the implications of projection choice.

Objectives

1. Explain the principles of geographic vs. projected coordinate systems.
2. Implement CRS transformations in R (or Python).
3. Quantify and visualize projection-induced distortions.
4. Evaluate the appropriateness of projections for specific analytical tasks.
5. Communicate findings clearly using reproducible code and visualizations.

Data

1. Use a Census polygon that includes multiple geographic regions:
US states, Texas counties, or Census tracts of a specific county.
2. The data must use a geographic CRS (e.g., WGS84/EPSG:4326).

Tasks

Part 1: Conceptual Background

Provide concise written responses to the following questions:

1. What is the difference between a *geographic coordinate reference system* (CRS) and a *projected coordinate reference system*?
2. Define the following types of projection distortion:
Area, Distance, Shape, Direction.
3. Explain why no single map projection can preserve all four properties simultaneously.

Part 2: Projection Selection and Transformation

Search/select **at least three** map projections representing different preservation goals (e.g., equal-area, conformal, equidistant, or compromise projections).

For each selected projection:

1. Transform the dataset from a geographic CRS into the target projected CRS.
2. Generate a map visualization of the transformed data.
3. Briefly describe the primary analytical purpose of the projection and the properties it is designed to preserve.

Part 3: Quantifying Projection Distortion

Using a consistent set of spatial features across all projections, complete the following analyses:

1. **Area Distortion**
 - Compute polygon areas under each selected projection.
 - Compare projected areas to a reference equal-area projection.
2. **Distance Distortion**
 - Select at least three meaningful point-to-point distances.
 - Compute distances under each projection and compare results.

3. Shape Distortion (Qualitative)

- Visually assess deformation of known regions or geometric features.
- Comment on angular distortion and shape preservation where relevant.

Summarize quantitative results using tables (and figures, or both).

Part 4: Visualization and Comparison

Create a comparative visualization that displays the same spatial dataset under each selected projection. Ensure that:

- Visual styling, symbology are consistent across projections where possible.
- Each map includes a clear title and CRS label.

Explain how observed visual differences correspond to the quantitative distortion measures computed in Part 3.

Part 5: Interpretation and Recommendation

Provide written responses to the following:

1. Which projection would you recommend for:
 - Global area comparison?
 - Navigation or bearing analysis?
 - Regional distance measurement?
2. What types of analytical errors can result from using an inappropriate projection?
3. How should projection choices be documented in professional or reproducible geospatial data workflows?

Files to be Submitted

You can submit the homework in two ways:

1. Original Jupyter Notebook that contains the output (no PDF or other format)
2. R script (or Python) + output file(s)

The basic *requirement* is that the I can run your submission and compare your submitted results against mine.